

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour

Total: Marks:40

Annexure A

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Code of Conduct:

I S. Chandra Shekhar (192372163) (Name / Reg No) certify that this submission is my original work and that I have adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Chandra Shekhar. S

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Industry Scenario

A healthcare company has patient records containing age, blood pressure, cholesterol, glucose level, BMI and family history. The objective is to predict diabetes early and recommend personalised treatment actions (Diet, Exercise, Medication, Monitor) based on patient-state transitions over time.

Task 1: Data Preparation & Inductive Learning

1(a) Inductive learning approach

Inductive learning learns a general predictive rule from labelled examples and applies it to unseen patients. Here, each patient record is an instance and the diabetes diagnosis is the target label. A suitable pipeline is: data cleaning → exploratory analysis → preprocessing → train/test split → model training → validation → performance evaluation.

Target variable: Diabetes (for example, 1 = diabetic and 0 = non-diabetic).

Key features:

- Glucose level – usually a strong indicator of abnormal blood-sugar regulation and therefore highly informative for diabetes prediction.
- BMI – represents body-weight status and is useful because higher BMI can be associated with increased diabetes risk.
- Age – diabetes risk can vary substantially with age and therefore provides useful predictive information.
- Blood pressure and cholesterol – useful additional cardiovascular/metabolic indicators.
- Family history – captures inherited or familial risk information.

1(b) Handling class imbalance

Because the dataset contains more non-diabetic than diabetic cases, accuracy alone can be misleading. I would first use a stratified train/test split and inspect the class distribution. For the training set, SMOTE is a suitable primary strategy because it creates synthetic minority-class examples and helps the classifier learn diabetic cases without simply duplicating records. Class weights can also be used as a comparison. SMOTE must be applied only to the training data to avoid data leakage.

1(c) 70:30 split and preprocessing

A 70:30 train/test split gives enough data for learning while retaining a substantial independent test set. In a medical application, the test set must remain unseen until final evaluation so that reported performance reflects generalisation.

- Numerical features: handle missing values using an appropriate training-set statistic, then standardise/normalise when required.
- Categorical feature such as family-history status: encode it numerically (e.g., 0/1).
- Target variable: encode diabetic/non-diabetic as 1/0.
- Use stratification so the diabetic/non-diabetic proportion is preserved in both sets.
- Fit preprocessing transformations on the training set only, then apply the learned transformation to the test set.

Task 2: Decision Tree for Diagnosis

2(a) Decision Tree and first three levels

A Decision Tree recursively selects the feature that gives the best class separation. The root should be the feature with the highest Information Gain or the lowest weighted Gini impurity after the split.

Illustrative three-level structure (the exact feature order must be obtained from the supplied dataset):

ROOT: Glucose ■■■ High glucose → likely Diabetic ■■■ Normal glucose → BMI ■■■ High BMI → Age ■ ■■■ Older → higher diabetes risk ■ ■■■ Younger → lower diabetes risk ■■■ Normal BMI → lower diabetes risk

Information Gain: $IG(S,A) = H(S) - \sum_v (|S_v|/|S|)H(S_v)$. Gini impurity: $Gini(S) = 1 - \sum_k p_k^2$. The root is selected using the feature that produces the greatest reduction in impurity. Exact Information Gain/Gini values require the actual dataset; they should not be invented.

2(b) Evaluation and False Negatives

The model should be evaluated using Accuracy, Precision, Recall and F1-score. In medical screening, Recall for the diabetic class is especially important because a False Negative means a patient with diabetes is predicted as non-diabetic.

Formulas: Accuracy = $(TP+TN)/(TP+TN+FP+FN)$; Precision = $TP/(TP+FP)$; Recall = $TP/(TP+FN)$; F1 = $2(Precision \times Recall)/(Precision+Recall)$.

To reduce False Negatives, use class weighting or an imbalance strategy such as SMOTE, tune the decision threshold to favour sensitivity, and validate using recall, F1-score and ROC-AUC. Clinically, missing a diabetic patient may delay diagnosis and treatment, so higher sensitivity is often prioritised for a screening system.

2(c) Post-pruning

Pre-pruning limits tree growth during training using parameters such as maximum depth, minimum samples per split or minimum samples per leaf. Post-pruning first grows a larger tree and then removes branches using cost-complexity pruning.

Cost-complexity objective: $R_\alpha(T) = R(T) + \alpha|T|$, where $R(T)$ is the tree's error measure, $|T|$ is the number of terminal nodes and α controls the complexity penalty.

The pruned model should be selected using validation performance and clinical priorities. A slightly less accurate but simpler tree with better generalisation, fewer unnecessary branches and improved interpretability is generally preferable for clinical deployment.

Comparison table:

Task 3: Statistical Learning for Risk Stratification

3(a) Logistic Regression

Logistic Regression is suitable because the target is binary. It models the probability of diabetes using a logistic function: $P(Y=1|X) = 1/(1+e^{-z})$, where $z = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$.

Compare the Decision Tree and Logistic Regression using Accuracy and AUC-ROC on the same untouched test set. AUC-ROC measures how well the model ranks positive cases across classification thresholds. Exact numerical Accuracy and AUC-ROC cannot be calculated from the assignment sheet alone because the patient dataset and model predictions are not provided.

A statistical model is preferable when the relationship is reasonably smooth/linear in the predictors, interpretability of coefficients is important, probability estimates are required, and a simpler model is desirable.

3(b) Feature importance

For the Decision Tree, feature importance can be calculated from the total impurity reduction contributed by each feature. For Logistic Regression, standardised coefficients can be compared by magnitude and direction. The top three predictors should be reported after training.

Likely candidates include glucose, BMI and age, but the final top-three ranking must be calculated from the actual dataset. The two models may disagree because a tree captures nonlinear threshold effects and interactions, while Logistic Regression estimates additive linear effects unless interactions or nonlinear transformations are explicitly added.

Task 4: Reinforcement Learning for Treatment Recommendation

4(a) MDP formulation

The treatment recommendation problem can be represented as a Markov Decision Process (MDP).

4(b) Q-Learning

Q-Learning updates the action-value function using: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$. Here α is the learning rate and γ is the discount factor.

For demonstration, use $\alpha=0.5$ and $\gamma=0.9$. The following is an illustrative Q-table update; actual clinical values must be generated from a defined transition/reward dataset.

Train for at least five episodes using an exploration policy such as ϵ -greedy. A practical convergence criterion is that the maximum absolute change in Q-values remains below a small threshold (for example, 0.01) over several consecutive episodes, together with stable policy selection.

4(c) Final policy and ethics

The learned policy is obtained by selecting the action with the highest Q-value in each state: $\pi(s)=\operatorname{argmax}_a Q(s,a)$. An illustrative policy is S0→Monitor, S1→Diet, S2→Exercise/Diet, and S3→clinician-supervised Medication/Monitor. The final policy must be determined from the trained Q-table and must not automatically prescribe medication.

- Patient safety: the agent must never override clinical judgement and should include safety constraints and escalation to a qualified clinician.
- Bias: monitor performance across demographic and clinical subgroups and check whether the training data under-represents any group.
- Explainability: show why an action was recommended, the important patient factors, confidence/uncertainty and relevant rules.
- Privacy: protect patient records and use appropriate access controls, anonymisation and secure storage.
- Human oversight: medication recommendations should require clinician review and approval.
- Monitoring after deployment: continuously audit errors, drift, adverse outcomes and subgroup performance.

Conclusion

The proposed solution combines inductive classification, a pruned Decision Tree, statistical risk modelling and a constrained Reinforcement Learning framework. For a clinical system, reducing missed diabetes cases, maintaining interpretability, validating on unseen data and enforcing human oversight are more important than maximising accuracy alone.

Important note on calculations

The assessment sheet specifies the required analyses but does not include the patient dataset or prediction outputs. Therefore, exact Information Gain/Gini values, Accuracy, Precision, Recall, F1, AUC-ROC, learned feature rankings and a genuine five-episode Q-table cannot be truthfully calculated from the sheet alone. The worked Q-learning values above are explicitly illustrative. They should be replaced with values obtained from the actual dataset before submission.

Model	Expected behaviour	Deployment view
Pre-pruned tree	Controlled complexity; may avoid overfitting	Useful baseline
Post-pruned tree	Removes weak branches and improves simplicity	Usually preferable if validation performance is maintained/improved

health stages such as S0 = controlled/low risk, S1 = elevated risk, S2 = high risk, S3 = diagnosed/poor control, with state variables derived from glucose, BMI, blood pressure and Medication, Monitor.

for measurable health improvement and adherence; negative reward for deterioration, adverse outcomes or unsafe actions. Medication decisions must be constrained by clinical rules. The reward depends on the current patient state, selected action and the patient's response over time. Transition probabilities can be estimated from longitudinal records or simulated for a teaching purpose.

Iteration	State	Action	Reward / Next State	Updated Q
1	S1	Diet	+2 → S0	$Q_{\text{new}} = 0 + 0.5[2 + 0.9(0) - 0] = 1.00$
1	S2	Exercise	+2 → S1	$Q_{\text{new}} = 0 + 0.5[2 + 0.9(0) - 0] = 1.00$
1	S2	Medication	+3 → S1	$Q_{\text{new}} = 0 + 0.5[3 + 0.9(0) - 0] = 1.50$
2	S1	Diet	+2 → S0	$Q_{\text{new}} = 1 + 0.5[2 + 0.9(0) - 1] = 1.50$
2	S2	Exercise	+2 → S1	$Q_{\text{new}} = 1 + 0.5[2 + 0.9(1.50) - 1] = 1.675$
2	S2	Medication	+3 → S1	$Q_{\text{new}} = 1.50 + 0.5[3 + 0.9(1.50) - 1.50] = 2.925$